

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology and Engineering

Devang Patel Institute of Advance Technology and Research (DEPSTAR)

**Department of Information Technology ITUC301:
Data Engineering**

PRACTICAL - 1

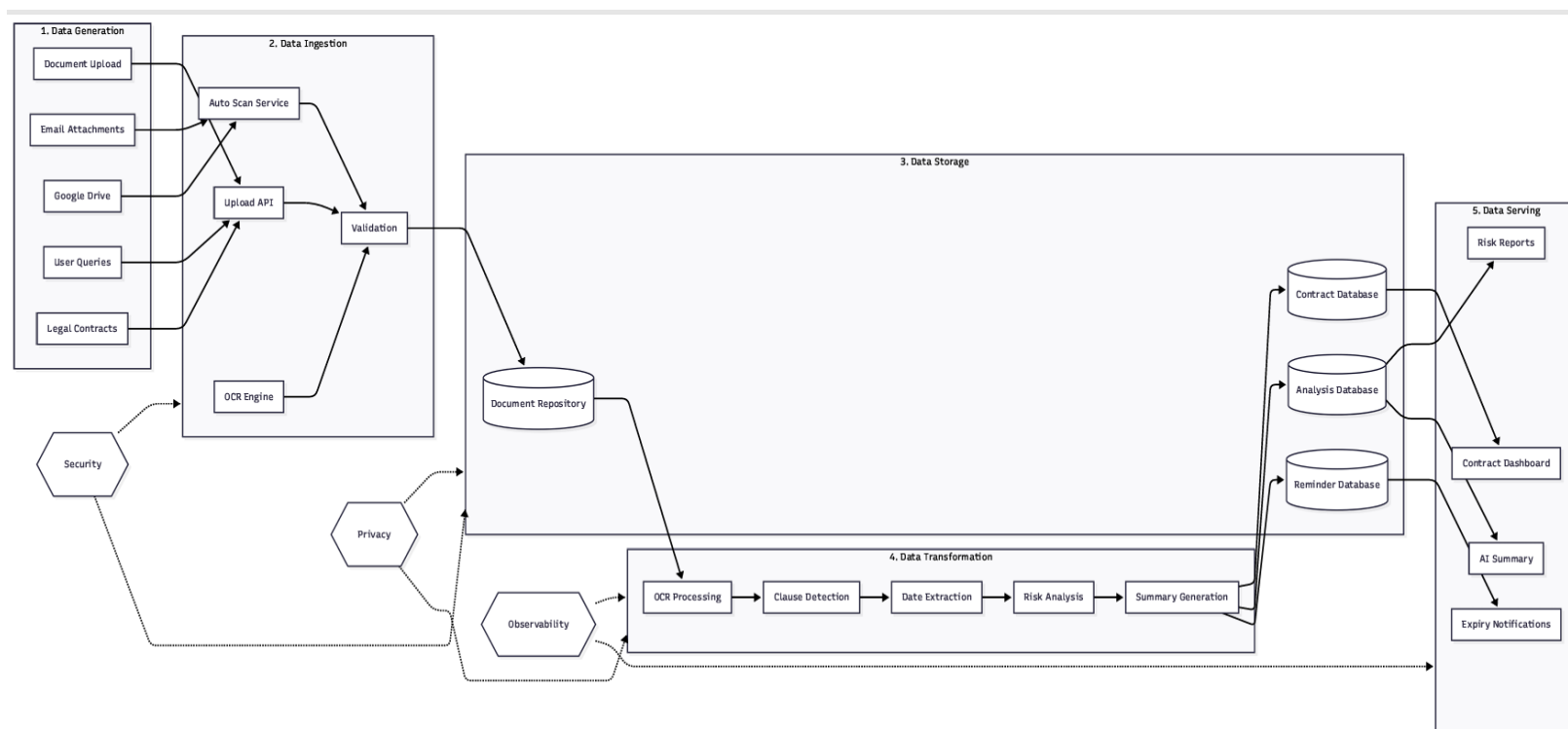
Title : Understanding the Data Engineering Lifecycle

Problem Definition: Study, dissect, and map the complete end-to-end data engineering lifecycle for an enterprise e-commerce platform. Students must trace data states across all five core lifecycle stages: data generation, ingestion, storage, transformation, and serving.

The solution requires creating a detailed architectural data flow diagram along with a system manifesto documenting cross-cutting lifecycle undercurrents (security, observability, and data privacy). This initial structure must be committed to a personal Git repository.

Dataset/Test Data: Mock enterprise e-commerce dataset schema definition (containing transactional database entities, clickstream tracking JSON payloads, and third-party inventory log streams).

OUTPUT :



Key Questions :**1: How do the technical demands of downstream serving layers directly influence ingestion and storage design decisions during the early generation phases?**

- Downstream serving needs (e.g. real-time recommendations, live dashboards) determine whether ingestion must be streaming vs batch, and whether storage should prioritize fast query access (warehouse) or cheap bulk storage (data lake). These serving-layer demands are considered even during generation, so the right data format and granularity is captured from the start.

2: Differentiate between the broader data lifecycle and the technical data engineering lifecycle within an enterprise.

- The broader data lifecycle covers data's entire journey across an organization — creation, business use, governance, archival, and deletion. The technical data engineering lifecycle is the narrower subset engineers directly build and operate: ingestion, storage, transformation, and serving pipelines.

3: Identify how the undercurrents of security and data observability should be enforced at each lifecycle boundary.

- Security is enforced at each boundary through encryption in transit/at rest and role-based access control before data moves to the next stage. Observability is enforced through logging, monitoring, and data quality checks at every transition, so failures or data drift are caught early rather than downstream.

Supplementary Problems

Problem 1: Smart City IoT Traffic Sensor Platform

Design the complete data engineering lifecycle for a smart city traffic monitoring system where thousands of IoT sensors continuously generate real-time traffic data. Explain how the data flows through the stages of generation, ingestion, storage, transformation, and serving. Highlight challenges such as high data velocity, real-time processing, scalability, and fault tolerance.

Solution (Short):

- **Generation:** Traffic sensors continuously generate vehicle count, speed, GPS location, and timestamp data.
- **Ingestion:** Apache Kafka or MQTT collects streaming sensor data in real time.
- **Storage:** Raw data is stored in a Data Lake, while processed data is maintained in a Data Warehouse for analytics.
- **Transformation:** Spark Streaming or Flink cleans, filters, aggregates, and enriches the data.
- **Serving:** Dashboards display live traffic conditions, congestion alerts, and route optimization suggestions.

Challenges:

- Continuous high-volume data streams
- Low-latency processing requirements
- Sensor failures and missing data
- Data privacy and secure transmission
- Scalable storage for long-term historical analysis

Problem 2: Online Food Delivery Platform

Map the data engineering lifecycle for an online food delivery application.

Solution (Short):

- Customer places an order through the mobile app.
- Order information is ingested into the platform.
- Data is stored in transactional databases and data lakes.
- Transformation generates sales reports and delivery analytics.
- Processed data is served through dashboards and recommendation systems.

Conclusion

The Data Engineering Lifecycle provides a structured approach for collecting, processing, storing, and serving data efficiently. In this practical, the lifecycle of an enterprise e-commerce platform was analyzed through the five major stages: **Data Generation, Data Ingestion, Data Storage, Data Transformation, and Data Serving**. Cross-cutting concerns such as **security, observability, and data privacy** were also considered throughout the lifecycle. Understanding these stages helps in designing reliable, scalable, and secure data pipelines that support business intelligence and data-driven decision-making. The practical also demonstrated the importance of documentation, architectural planning, and version control using Git for professional data engineering projects.